Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/22/2021
PI Name:	Rosi, Susanna Ph.D.		
Project Title:	The Role of Oxidative Stress and Inflammation on Synaptic Functions After Exposure to Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94110-3518	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	01/06/2014	**End Date:**	06/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 4/26/21) NOTE: End date changed to 1/31/2021 per NSSC information (Ed., 7/23/2020) NOTE: End date changed to 6/30/2020 per NSSC information (Ed., 1/29/2020) NOTE: End date changed to 1/5/2020 per NSSC information (Ed., 2/12/19) NOTE: End date changed to 1/5/2019 per S. Monk/HRP-LaRC (Ed., 12/21/17)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Nelson, Gregory Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX14AC94G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Radiation exposure significantly affects the hippocampal formation, a structure critical for cognitive function. We demonstrated that low doses of 56Fe irradiation induced changes in cognition that correlated with changes in the expression of memory-related proteins and that low doses of space radiation and oxidative stress were associated with reductions in dendritic complexity and spine density. The underlying mechanisms for radiation-induced cognitive impairments are unknown, but likely are multifaceted, involving inflammation and chronic oxidative stress. Persistent oxidative stress has been demonstrated in space radiation studies in animal models; our preliminary data demonstrate that low doses of space radiation alter the expression of inflammatory chemokines and the dendritic system in the hippocampus. Maintenance of the dendritic system, including spine density and dendritic arbors, is essential for proper synaptic functions. Therefore, the dendritic system may be the missing link between oxidative stress/neuroinflammation and cognitive decline following irradiation. The purpose of this application is to determine how space-relevant doses of radiation affect dendritic maintenance as it pertains to synaptic functions and memory. We hypothesize that space radiation, even at low doses, impairs synaptic functions with negative consequences for cognition and that these effects are mediated by oxidative stress and inflammation. We will use novel mouse models with neuronal-specific overexpression of an antioxidant enzyme to address the role of oxidative stress. The mice will be exposed to space-relevant doses of proton, Helium, Oxygen, or a combination of the three and analyzed for late effects. Experimental endpoints will include cognitive functions, dendritic structures and synaptic functions, molecules that control the dendritic system, inflammatory responses, and indices of oxidative stress. This systematic approach will allow us: 1) to connect the functional output of cognition to the molecular pathway that controls synaptic plasticity and memory; and 2) to determine the role of oxidative stress and inflammation in space radiation-mediated changes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed aims directly address NASA's Human Research Program (HRP) Integrated Research Plan (IRP) program elements that specify a number of gaps related to the acute and late risks to the central nervous system (CNS). Specific gaps addressed by this proposal are the acute functional changes and the long-term CNS risk and the mechanisms of changes. Data obtained from our mouse models will also help to estimate late CNS risks to astronauts. On a broader scope, the use of mouse models to address the role of oxidative stress and inflammation in space radiation-mediated CNS risk should be beneficial in filling in the gaps of degenerative risks to the extent they apply to the CNS.
Task Progress:	Here, we cement the essential role of microglia activation in space-like radiation (GCRsim)-induced cognitive decline in male mice. Furthermore, we identify a therapeutic (PLX5622) for mitigation of the radiation-induced spatial learning deficits. Finally, we found that peripheral blood monocyte levels can be used as an early biomarker for predicting long-term GCRsim-mediated microglia activation and cognitive decline in male mice. In line with our previous study that used different ion composition exposure, a sex-dimorphic response was measured: all deficits were observed in male mice, with female mice protected from GCRsim-induced phenotypes.
Bibliography Type:	Description: (Last Updated: 09/04/2023)
Articles in Peer-reviewed Journals	Krukowski K, Grue K, Becker M, Elizarraras E, Frias E, Halvorsen A, Koenig-Zanoff M, Frattini V, Nimmagadda H, Feng X, Jones T, Nelson G, Ferguson AR, Rosi S. "The impact of deep space radiation on cognitive performance: From biological sex to biomarkers to countermeasures." Sci Adv. 2021 Oct 15;7(42):eabg6702. https://doi.org/10.1126/sciadv.abg6702 ; PMID: 34652936; PMCID: PMC8519563 , Oct-2021
Articles in Peer-reviewed Journals	Paladini MS, Feng X, Krukowski K, Rosi S. "Microglia depletion and cognitive functions after brain injury: From trauma to galactic cosmic ray." Neurosci Lett. 2021 Jan 10;741:135462. Epub 2020 Nov 28. https://doi.org/10.1016/j.neulet.2020.135462 ; PMID: 33259927 , Jan-2021
Articles in Peer-reviewed Journals	Rienecker KDA, Paladini MS, Grue K, Krukowski K, Rosi S. "Microglia: Ally and enemy in deep space." Neurosci Biobehav Rev. 2021 Jul;126:509-14. Epub 2021 Apr 16. https://doi.org/10.1016/j.neubiorev.2021.03.036 ; PMID: 33862064 , Jul-2021
Articles in Peer-reviewed Journals	Borlongan MC, Rosi S. "Stem cell therapy for sequestration of traumatic brain injury-induced inflammation." Int J Mol Sci. 2022 Sep 7;23(18):10286. https://doi.org/10.3390/ijms231810286 ; PubMed PMID: 36142198; PubMed Central PMCID: PMC9499317 , Sep-2022